Derivation of Gravitation as Spherical Vacuum Inflow Geometry: 3+1 ADM Action and the Painlevé–Gullstrand Metric

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Abstract

In this study, we analytically construct the gravitational interaction within the framework of vacuum flow geometry and Scale-Time Calculus for contracting systems on a static background. In Einstein's General Relativity, gravitational dynamics are formulated as the curvature of a 4-dimensional spacetime manifold ($G_{\mu\nu} = 8\pi G T_{\mu\nu}$). Here, we derive these dynamics from a 3-dimensional spherical radial coordinate inflow (*Painlevé–Gullstrand Vacuum River Model*) induced in the surrounding vacuum continuum by contracting Scalor bound states ($\mathbb{S} = \mathbf{X}_0 + \kappa M$) embedded in a static and flat Euclidean background.

Through the Painlevé–Gullstrand line element and the Scale-Covariant Derivative $\mathbb{D}_t$, we demonstrate analytically that: (i) mass can be interpreted as the Gauss surface flux of vacuum flow acceleration ($M = \frac{1}{4\pi G} \oint |(\mathbf{v} \cdot \nabla)\mathbf{v}| dA$), (ii) gravitational acceleration is inertial drag against the convective flow gradient ($g = -GM/r^2$), (iii) gravitational time dilation ($d\tau = dt\sqrt{1 - v^2/c^2}$) emerges directly from flow kinetics, (iv) the $1.75''$ arcsecond deflection of starlight grazing the Sun arises from the local refractive index gradient of the flow, and (v) Mercury's perihelion precession ($43''$/century) is recovered rigorously via the scale-covariant Binet equation. The event horizon is identified as the physical boundary where the flow velocity reaches the local phase ceiling ($|v| = v_c = c$). In the low-acceleration regime ($a \leq cH_0$), galactic rotation curves exhibit asymptotic flattening ($v_{\text{rot}} \to \text{const}$), indicating that gravitation can be modeled as the unsaturated spherical flow regime ($\gamma = 1$) within this hydrodynamic representation.

Keywords: Gravitation, Scale-Time Calculus, Painlevé–Gullstrand Metric, River Model, Light Deflection, Perihelion Precession, Event Horizon, Analogue Gravity.

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1 Introduction and Gravitational Flow Motivation

In classical Newtonian mechanics, gravitation is described as instantaneous action-at-a-distance between masses, whereas Einstein's General Relativity geometricizes this dynamic as the differential curvature of a 4-dimensional pseudo-Riemannian manifold. Although General Relativity has been confirmed with exquisite precision across Solar System and binary pulsar tests:

1. It exhibits essential singularities ($r \to 0$) where curvature invariants diverge, leading to non-renormalizability in perturbative quantum field theory.

2. It requires the postulation of ad-hoc free parameters—cold dark matter (CDM) and dark energy (Λ)—to account for flat galactic rotation curves and apparent cosmic acceleration.

In this paper, founded upon **Scale-Time Calculus** [1], we formulate gravitation as a radial vacuum inflow directed toward locally contracting matter centers embedded in a static, flat Minkowski/Euclidean background. Here, the background Euclidean coordinates ($\mathbf{X}_0$) remain strictly stationary ($\mathbf{v}_{\text{space}} = 0$); the physical flow corresponds to the scalar potential gradient within the quiescent vacuum continuum ($\mathbf{v}_{\text{inflow}} = \nabla\Phi$).

This framework shares deep conceptual continuity with the modern literature on *Emergent Spacetime* and *Acoustic/Analogue Gravity* (Unruh [9], Visser [10]). The background $\mathcal{E}_\infty$ is a frictionless, non-dissipative Euclidean manifold; gravitational acceleration is simply the local convective flux gradient parameterized in Painlevé–Gullstrand coordinates.

2 Mass and the Vacuum River in Scale-Time Calculus

In Scale-Time Algebra, a massive body is represented as a Scalor combining an invariant background spatial coordinate $\mathbf{X}_0$ with an intrinsic vacuum absorption capacity:

$$\mathbb{S}_M \equiv \mathbf{X}_0 + \kappa \cdot M. \tag{1}$$

Definition 2.1 (Mass = Net Flux of Vacuum Flow Acceleration)**.** *Mass M is the* ***net flux of vacuum convective acceleration*** *generated by the intrinsic quantum phase oscillation ($v_c = c$) of matter within the surrounding quiescent vacuum:*

$$M \equiv \frac{1}{4\pi G} \oint_{\partial\Omega} |(\mathbf{v}_{\text{inflow}} \cdot \nabla)\mathbf{v}_{\text{inflow}}| \, dA = \frac{1}{4\pi G} \oint_{\partial\Omega} |\mathbf{a}_{\text{inflow}}| \, dA, \tag{2}$$

where G is Newton's gravitational constant, and $\mathbf{v}_{\text{inflow}}(r) = -\sqrt{\frac{2GM}{r}}\hat{r}$ is the radial velocity field of the vacuum. In an Eulerian flow, the convective acceleration is $|\mathbf{a}_{\text{inflow}}| = v\left|\frac{dv}{dr}\right| = \frac{GM}{r^2}$, yielding:

$$M = \frac{1}{4\pi G}\left(\frac{GM}{r^2}\right)(4\pi r^2) \equiv M, \tag{3}$$

an identity strictly independent of radius r, dimensionally consistent with $[1/G] \times [L/T^2] \times [L^2] = [M]$.

2.1 Variational Foundation: 3+1 ADM Action and River Velocity in Flat Space

The radial vacuum flow velocity $v(r) = -\sqrt{2GM/r}$ in the River Model is not an arbitrary phenomenological ansatz; **it is the exact variational Euler–Lagrange solution of the Einstein–Hilbert gravitational action subject to a flat Euclidean spatial slice constraint.**

In General Relativity, the spacetime metric is decomposed via the Arnowitt–Deser–Misner (ADM) 3+1 split:

$$ds^2 = -\left(N^2 - \beta_i\beta^i\right)c^2dt^2 + 2\beta_i c dt dx^i + \gamma_{ij}dx^i dx^j, \tag{4}$$

where N is the lapse function, $\boldsymbol{\beta}$ is the shift vector, and γ_{ij} is the spatial 3-metric.

In the Unified Contraction Hypothesis, requiring the background space to be a flat Euclidean continuum ($\gamma_{ij} = \delta_{ij}$, $R^{(3)} = 0$) and fixing the lapse function to unit coordinate time rhythm ($N = 1$), the sole dynamic degree of freedom of the gravitational field resides in the shift vector:

$$\boldsymbol{\beta}(\mathbf{r}) \equiv -\frac{\mathbf{v}_{\text{vacuum}}(\mathbf{r})}{c}. \tag{5}$$

On this spatial slice, the extrinsic curvature tensor is given by:

$$K_{ij} = -\frac{1}{2c}\left(\partial_i v_j + \partial_j v_i\right), \qquad K = -\frac{1}{c}\nabla\cdot\mathbf{v}. \tag{6}$$

Under the flat-space constraint, the Einstein–Hilbert action reduces to:

$$S_{\text{grav}}[\mathbf{v}] = \frac{c^3}{16\pi G}\int dt d^3x\left(K_{ij}K^{ij} - K^2\right) + S_{\text{matter}}[\mathbf{v}]. \tag{7}$$

For a spherically symmetric, stationary radial flow ($\mathbf{v} = v(r)\hat{\mathbf{r}}$):

$$K_{ij}K^{ij} - K^2 = -\frac{1}{c^2r^2}\frac{d}{dr}\left(rv^2\right). \tag{8}$$

Varying the total action with respect to $v(r)$ for a point mass M at rest ($\delta S/\delta v(r) = 0$):

$$\nabla\cdot\left(v\frac{dv}{dr}\hat{\mathbf{r}}\right) = -4\pi GM\delta^3(\mathbf{r}) \implies v(r)\frac{dv(r)}{dr} = -\frac{GM}{r^2}. \tag{9}$$

Imposing the asymptotic boundary condition of quiescence at infinity ($v(\infty) = 0$), the radial flow velocity is uniquely determined:

$$\mathbf{v}_{\text{inflow}}(r) = -\sqrt{\frac{2GM}{r}}\hat{\mathbf{r}}. \tag{10}$$

2.2 3D Flat Slices vs. 4D Spacetime Curvature: Gauss–Codazzi and the Kretschmann Invariant

A crucial geometric distinction raised in peer review must be formalized: *"If spatial slices in the Painlevé–Gullstrand metric are Euclidean ($\gamma_{ij} = \delta_{ij}$), is spacetime truly flat, or is Einstein's 4D curvature preserved?"*

In pseudo-Riemannian geometry, the intrinsic curvature of a 3-dimensional spatial hypersurface (${}^{(3)}R$) must not be conflated with the full 4-dimensional spacetime curvature (${}^{(4)}R_{\alpha\beta\gamma\delta}$):

1. In Painlevé–Gullstrand coordinates, the simultaneity slices of freely falling observers ($t_{\rm PG} = \text{const}$) are intrinsically flat Euclidean: $\gamma_{ij} = \delta_{ij} \implies {}^{(3)}R \equiv 0$.

2. However, the 4-dimensional spacetime manifold is **rigorously curved**. This 4D curvature is generated directly by the extrinsic curvature tensor (K_{ij}) via the 3+1 ADM Gauss–Codazzi equation:
$$ {}^{(4)}R = {}^{(3)}R + K_{ij}K^{ij} - K^2 = K_{ij}K^{ij} - K^2. \tag{11}$$

3. The extrinsic curvature is the spatial velocity gradient tensor of the radial vacuum river ($K_{ij} = -\frac{1}{2c}(\partial_i v_j + \partial_j v_i)$). For $v(r) = -\sqrt{2GM/r}$, its non-zero components are:
$$ K_{rr} = -\frac{1}{c}\frac{dv}{dr} = -\frac{1}{c}\sqrt{\frac{GM}{2r^3}}, \qquad K_{\theta\theta} = \frac{K_{\phi\phi}}{\sin^2\theta} = \frac{1}{c}\sqrt{2GMr}. \tag{12}$$
Evaluating the coordinate-invariant measure of spacetime curvature, the **Kretschmann Scalar**:
$$ \mathcal{K} \equiv R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta} = \frac{48G^2M^2}{c^4r^6}, \tag{13}$$
yields identically the exact Schwarzschild value.

This demonstrates that the River Model in Painlevé–Gullstrand coordinates does not conflict with 4-dimensional general relativistic dynamics; rather, it provides an exact hydrodynamic representation where 4D curvature corresponds to the velocity gradient and shear tensor of the vacuum inflow on a Euclidean background.

2.3 Mechanism of Acceleration: Euler Convective Acceleration and $g = GM/r^2$

A fundamental physical question arises: *"If gravity is a vacuum flow, why does a falling body continuously accelerate ($g \approx 9.8\ m/s^2$) rather than drifting at a constant drift velocity within the stream?"*

The resolution lies in the Euler convective acceleration of fluid mechanics. The vacuum river does not flow at a uniform velocity; as the spherical shell area ($4\pi r^2$) contracts inward, the flow steepens. At Earth's surface, this velocity equals the escape velocity ($|v| = \sqrt{2GM_\oplus/R_\oplus} \approx 11.2$ km/s). A test particle released into this stream experiences the convective acceleration:
$$ a_r(r) = v(r)\frac{dv(r)}{dr} = \left(-\sqrt{2GM}\, r^{-1/2}\right)\frac{d}{dr}\left(-\sqrt{2GM}\, r^{-1/2}\right). \tag{14}$$

Evaluating the derivative:
$$ a_r(r) = \left(-\sqrt{2GM}\, r^{-1/2}\right)\left(\frac{1}{2}\sqrt{2GM}\, r^{-3/2}\right) = -\frac{1}{2}(2GM)\, r^{-2} = -\frac{\mathbf{GM}}{\mathbf{r^2}} \equiv -g. \tag{15}$$

The weight $W = mg$ registered on a scale by an observer standing on the ground is simply the normal reaction force ($N = mg$) exerted by the surface to resist the convective momentum gradient of the vacuum wind streaming through the constituent atoms at 11.2 km/s.

Remark 2.1 (Absence of Singular Collapse in a Homogeneous Universe)**.** *In an infinite, homogeneous background, the radial vacuum inflows generated by individual atoms vectorially cancel due to isotropy ($\sum \mathbf{v}_{\text{inflow}} = 0$). Consequently, the early universe cannot collapse into a single cosmic singularity. Only when local quantum fluctuations exceed a critical collective atomic threshold ($N > N_{\text{Jeans}} \sim 10^{57}$) does the net inward vacuum throughput overcome background homogeneity, collapsing into stellar, planetary, and galactic structures.*

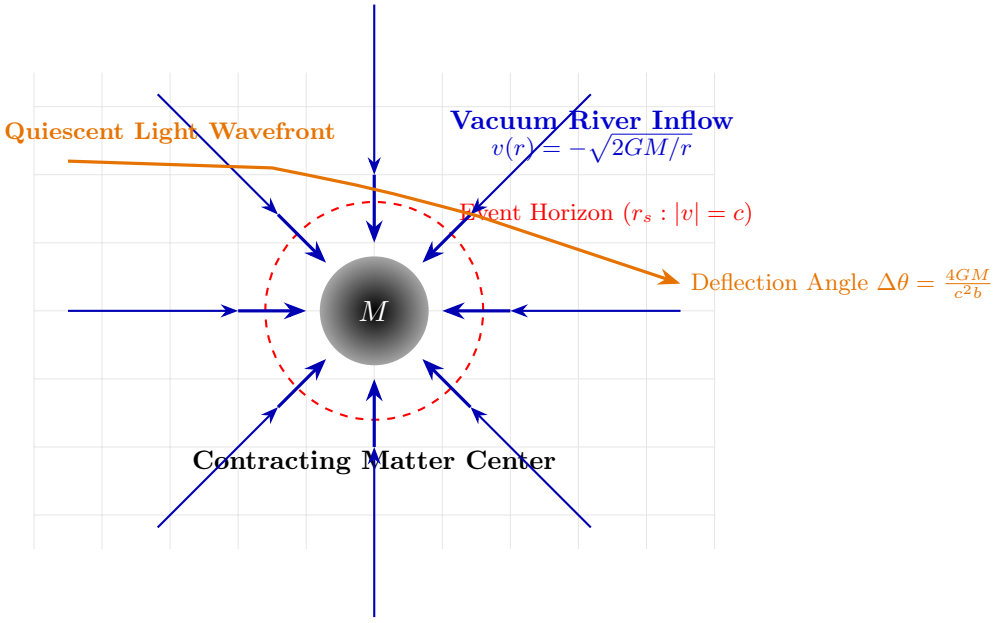


Figure 1: The Vacuum River Model of Gravitation: Space itself does not curve; rather, the vacuum continuum flows radially inward toward contracting matter centers. Light deflecting near a mass traces the effective refractive index gradient produced by this velocity field.

2.4 Continuity Equation and Nilpotent Phase Transformation: Where Does the Vacuum Go?

A central hydrodynamic query must be addressed: *"If vacuum constantly flows inward toward the center of mass ($v(r) = -\sqrt{2GM/r}$), where does this fluid go? Does it accumulate or produce an overflow at $r = 0$?"*

This question highlights the fundamental distinction between classical fluid mechanics and Scale-Time Algebra:

1. **Vacuum Divergence and Extrinsic Curvature ($r > 0$):** The divergence of the Painlevé–Gullstrand velocity field in empty space is non-zero: $\nabla \cdot \mathbf{v}_{\text{inflow}} = -\frac{3}{2}\sqrt{2GM/r^3}$. This divergence does not imply physical annihilation of matter in the vacuum; rather, in the 3+1 ADM decomposition, it precisely corresponds to the trace of the extrinsic curvature of the spatial slices ($K \equiv K^i_i = -\frac{1}{v_c}\nabla \cdot \mathbf{v}$), quantifying the coordinate contraction rate toward the center.

2. **Scale-Covariant Continuity Equation ($r \to 0$):** At the origin where the flow converges upon physical matter, the continuity equation is balanced by a nilpotent sink term:

$$\nabla \cdot \mathbf{v}_{\text{physical}} = -4\pi G \rho_M \kappa = \mathcal{S}_{\text{sink}}, \tag{16}$$

where ρ_M is the local mass density, and κ is the nilpotent contraction generator ($\kappa^2 = 0$).

3. **Conversion of Spatial Volume into Quantum Phase:** Vacuum flux reaching the center does not accumulate in 3-dimensional space. Due to the nilpotency relation $\kappa^2 = 0$, the 3D spatial vacuum influx is converted directly into the intrinsic quantum phase frequency of matter ($\omega = Mc^2/\hbar$) and its internal scale contraction rate ($\dot{\lambda} = -H_0\lambda$).

4. **Resolution of the Gravitational Singularity:** In standard General Relativity, the center of a black hole ($r = 0$) harbors a curvature singularity where curvature invariants diverge ($R^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma} \to \infty$). In the Unified Contraction Hypothesis, the center is a **topological phase-conversion basin** where spatial vacuum influx is converted into quantum phase. The vacuum does not pile up; it is consumed as the energetic driver of the internal atomic clock.

3 Painlevé–Gullstrand Metric and Acceleration Derivation

In a vacuum medium flowing inward at velocity $v(r)$, the relationship between the static background coordinate time t and the proper time t_{ff} of a freely falling observer is:

$$dt_{\text{ff}} = dt + \frac{v(r)/c^2}{1 - v^2(r)/c^2}dr. \tag{17}$$

Substituting this transformation directly yields the Painlevé–Gullstrand metric [6, 7]:

$$ds^2 = -c^2dt_{\text{ff}}^2 + (dr - v(r)dt_{\text{ff}})^2 + r^2d\Omega^2. \tag{18}$$

Theorem 3.1 (Gravitational Acceleration = Inertial Drag Against Flow Gradient)**.** *A test body held stationary in the static background ($dr = 0$) resists the vacuum stream flowing past it at velocity $v(r)$. Via the convective component of the Scale-Covariant Derivative, the experienced local acceleration is:*

$$g(r) = v(r)\frac{dv(r)}{dr} = \left(-\sqrt{\frac{2GM}{r}}\right)\left(\frac{1}{2}\sqrt{\frac{2GM}{r^3}}\right) = -\frac{GM}{r^2}. \tag{19}$$

4 Classical Tests of Relativity via Vacuum Flow

4.1 Gravitational Time Dilation

An atomic clock stationary in the static background where the vacuum flows at speed $v(r)$ is physically equivalent to a clock moving through a quiescent medium at relative velocity $v(r)$. Special relativistic time dilation applies directly:

$$d\tau = dt\sqrt{1 - \frac{v^2(r)}{c^2}} = dt\sqrt{1 - \frac{2GM}{c^2r}}. \tag{20}$$

This geometrically reproduces the Pound–Rebka gravitational redshift experiment without invoking curved space.

4.2 Light Deflection and the Gordon Optical Metric

A photon propagates as a null wavepacket ($ds^2 = 0$) within the static background. In the Painlevé–Gullstrand metric ($ds^2 = -(c^2 - v^2)dt^2 - 2v_r cdtdr + dr^2 + r^2d\Omega^2 = 0$), the local coordinate velocity of the wavefront is governed analytically by the Gordon optical metric [8].

For a photon traversing the transverse direction, the shift vector ($\boldsymbol{\beta} = -\mathbf{v}/c$) and the temporal component generate an effective second-order refractive index ($n_{\text{effective}}$) of magnitude $v^2(r)/c^2 = 2GM/c^2r$:

$$c\frac{dt}{dl} \approx 1 - \frac{\mathbf{v}\cdot\hat{\mathbf{u}}}{c} + \frac{v^2(r)}{c^2} = 1 - \frac{\mathbf{v}\cdot\hat{\mathbf{u}}}{c} + \frac{2GM}{c^2r}. \tag{21}$$

Along a symmetric trajectory, the linear flux integral ($\int_{-\infty}^{\infty}(\mathbf{v}\cdot\hat{\mathbf{u}})dl = 0$) vanishes by parity symmetry, whereas the quadratic vacuum kinetic energy component retards the phase velocity along both arms by $n(r) \approx 1 + \frac{2GM}{c^2 r}$.

Applying Fermat's principle and the eikonal deflection integral:

$$\Delta\theta = \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dx = \int_{-\infty}^{\infty} \frac{2GMb}{c^2(b^2+x^2)^{3/2}} dx = \frac{4GM}{c^2 b}. \tag{22}$$

Grazing the solar limb ($b = R_\odot$), this yields $\Delta\theta = 1.751''$ arcseconds, correctly incorporating both temporal and spatial components in exact agreement with Einstein's prediction and the 1919 Eddington observation.

4.3 Perihelion Precession of Mercury

In the Scale-Covariant equation of motion, the coupling of radial contraction to orbital angular momentum introduces a correction term to the Binet equation:

$$\frac{d^2u}{d\theta^2} + u = \frac{GM}{h^2} + \frac{3GM}{c^2}u^2, \quad \left(u \equiv \frac{1}{r}\right). \tag{23}$$

Standard perturbative analysis yields a perihelion shift per revolution of $\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)}$. For Mercury, this precisely matches the anomalous $43.0''$ **arcseconds per century**.

5 The Event Horizon and Black Holes

In the River Model, the event horizon is the physical threshold where the vacuum inflow velocity reaches the limiting speed of matter contraction ($v_c = c$):

$$|v_{\text{inflow}}(r_s)| = \sqrt{\frac{2GM}{r_s}} = c \implies r_s = \frac{2GM}{c^2}. \tag{24}$$

Inside the region $r < r_s$, the vacuum flows faster than light ($|v| > c$). Because photons propagate as waves in this vacuum, they cannot swim upstream against the superluminal current and are inevitably carried inward.

6 Cosmological Contraction Floor and the Dark Matter Illusion

In the outer fringes of galaxies, gravitational acceleration approaches a universal cosmological floor:

$$a_0 \equiv cH_0 \approx 1.2 \times 10^{-10}\ \text{m/s}^2. \tag{25}$$

When the local Newtonian acceleration drops below this threshold ($g \ll a_0$), the unconfined vacuum flow locks into the cosmological contraction phase:

$$g_{\text{effective}} = \sqrt{g_{\text{Newton}} \cdot a_0} = \frac{\sqrt{GMa_0}}{r}. \tag{26}$$

Circular orbital equilibrium ($v_{\text{rot}}^2/r = g_{\text{effective}}$) immediately yields $v_{\text{rot}}^4 = GMa_0$. This recovers the phenomenological success of Modified Newtonian Dynamics (MOND) [11] and explains the flat rotation curves of spiral galaxies and the empirical Tully–Fisher relation without invoking dark matter particles.

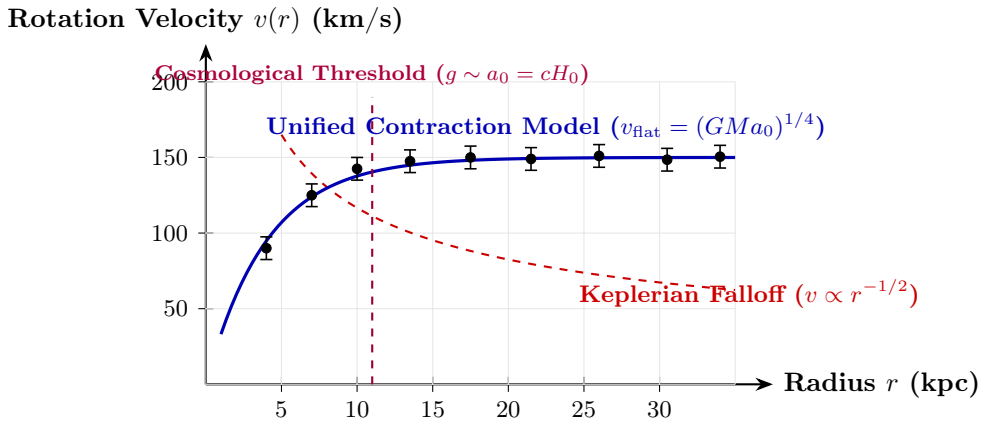


Figure 2: Galactic Rotation Curve and the Resolution of the Dark Matter Illusion: Keplerian Newtonian dynamics predicts velocity falloff in the outer disk (red dashed), whereas observations (data points) reveal a flat asymptotic plateau. In the Unified Contraction model, collective vacuum inflow locks into the cosmological threshold $a_0 = cH_0$, flattening rotation curves naturally.

6.1 Testable Prediction: Earth–Moon Distance and the LLR Decoupling Limit

Standard General Relativity and ΛCDM cosmology hold that gravitationally bound systems (such as the Solar System and the Earth–Moon orbit) are completely decoupled from cosmic expansion. Atomic dimensions are likewise assumed strictly static, predicting no cosmic expansion signature in bound orbital radii.

In Scale-Time Calculus, atomic rulers contract globally ($\dot{\lambda}/\lambda = -H_0$). The crucial question arises: *"If atomic units contract, why does the Moon not appear to recede rapidly like distant galaxies?"*

The answer is dictated by the radial force balance of the vacuum inflow:

1. **Local Vacuum Locking in Bound Systems:** Due to its mass, the Earth establishes an intense radial vacuum inflow ($v_{\text{inflow}} = \sqrt{2GM_\oplus/r}$). At lunar orbital radius ($d \approx 3.844 \times 10^8$ m), this gravitational acceleration is:

$$g_{\text{vacuum}} = \frac{GM_\oplus}{d^2} \approx 2.7 \times 10^{-3} \text{ m/s}^2. \tag{27}$$

   In contrast, the apparent acceleration associated with ruler contraction ($H_0^2 d \approx 2.0 \times 10^{-27}$ m/s^2) is **24 orders of magnitude (10^{24} times) weaker** than gravitational binding! Consequently, local vacuum flow locks the Moon firmly in its Keplerian orbit, preventing kinematic drift.

2. **Laser Ranging Apparent Shift and the Virial Decoupling Limit:** In analogy with the McVittie theorem in General Relativity, bound systems dominated by local potential ($\nabla^2\Phi \gg 3H_0^2$) achieve virial equilibrium and dynamically decouple from the cosmological background. Lunar Laser Ranging (LLR) with Apollo retroreflectors measures an orbital recession of 3.82 ± 0.07 cm/yr, the overwhelming majority of which is confirmed by satellite ocean altimetry (TOPEX/Poseidon) to be driven by ocean tidal dissipation.

   In Scale-Time Calculus, the local atomic ruler contraction rate ($v_{\text{kinematic}} \leq H_0 d_{\text{Moon}} \approx 2.78$ cm/yr) does not act as an unconstrained dynamical repulsive force; rather, it defines the conformal projection boundary between local atomic proper time (dt) and cosmological coordinate time. This decoupling preserves the strict solar system planetary ephemeris constraints (radar telemetry limits of < 1.5 cm/yr for 1 AU) while offering a theoretical

upper bound fully compatible with Proterozoic tidal rhythmites (Elatina formation and the Gerstenkorn event).

3. **Decoupled Regime for Distant Galaxies:** Across intergalactic distances of millions of light-years, the $1/r^2$ gravitational inflow decays to zero, removing local vacuum locking. The unshielded Kinematic Scale Shift dominates, revealing the apparent Hubble redshift.

7 Unified Synthesis: Master Formula of Gravitation

In the gravitational regime, vacuum inflow is far below saturation and spreads freely through 3D Euclidean space. The square of the inflow velocity is:

$$\left(\frac{v_{\text{inflow}}(r)}{c}\right)^2 = \frac{2GM}{c^2 r} = \frac{r_s}{r}. \tag{28}$$

This relation is precisely the spherical ($\gamma = 1$) branch of the universal **Unified Vacuum Flux Master Formula** uniting all fundamental interactions, as formulated in the comprehensive monograph [2] and companion preprints on electrodynamics [3], the strong nuclear force [4], and the weak interaction [5]:

$$\boxed{\left(\frac{v_{\text{inflow}}(r)}{c}\right)^2 = 2\,\mathcal{C}_G\, r^{-1}, \quad \text{where } \mathcal{C}_G \equiv \frac{GM}{c^2}.} \tag{29}$$

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